

Personal Expense Tracker MCP

A local MCP server for adding, reviewing, updating, deleting, and summarizing personal expenses.

Rafeef Sholy — Computer Engineer

3–5 MINUTES



The problem → a focused solution

Why we built the Personal Expense Tracker MCP

PROBLEM

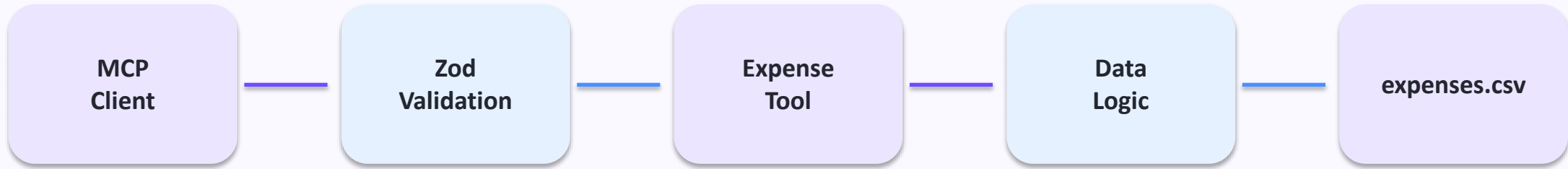
- Expense records can be scattered across notes and messages.
- Finding a specific expense or calculating totals manually is slow.
- An AI client should not receive unlimited access to files, commands, or the internet.

SOLUTION

- Five focused MCP tools handle the core expense actions:
`add_expense` • `list_expenses` • `get_spending_summary`
`update_expense` • `delete_expense`
- Inputs are validated before the expense logic runs.
- Data stays in one fixed local file: `data/expenses.csv`.

Architecture & project journey

How the system works, and how the project moved from data to a final demo-ready release



PROJECT JOURNEY

- Earlier weeks: planned the expense data, connected CSV storage, and built the MCP expense workflow.
- The project expanded into five validated tools for adding, listing, summarizing, updating, and deleting expenses.
- Final stage: tested from a fresh clone, verified build/run, prepared the demo, and published v1.0.0.

Live demo: exactly two prompts

The required live section focuses on a simple add → list flow

5 TOOLS

add_expense

list_expenses

get_spending_summary

update_expense

delete_expense

PROMPT 1

Add a \$25 groceries expense for milk and bread on August 16, 2026.

PROMPT 2

Show my groceries expenses for August 2026.

Backup prompt

Summarize August 2026 spending and show the category with the highest total.

Tool: [get_spending_summary](#)

Next steps & demo takeaway

Future improvements after the complete project journey

01

Encrypted backup

Optional backup while keeping the local-first design.

02

Better insights

Clearer spending charts and summaries for users.

03

Safe CSV import

Preview and validate imported data before writing.

Takeaway: focused MCP tools + strict validation + safe local storage + bounded permissions.

READY FOR QUESTIONS